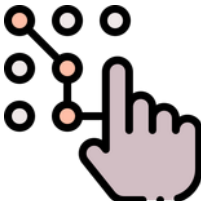


DYNAMIC PRICE OPTIMASATION FOR URBANMART

Jay | Diva | Rucha

Agenda

- Project goal and data overview and insights. 
- Insight 1 : Our top selling products and categories. 
- Insight 2 : When are customers most active? 
- Insight 3 : What are the common reorder patterns? 
- Insight 4 : Which products are bought together? 
- Business recommendations for FreshMart 

Project Goal & Data overview

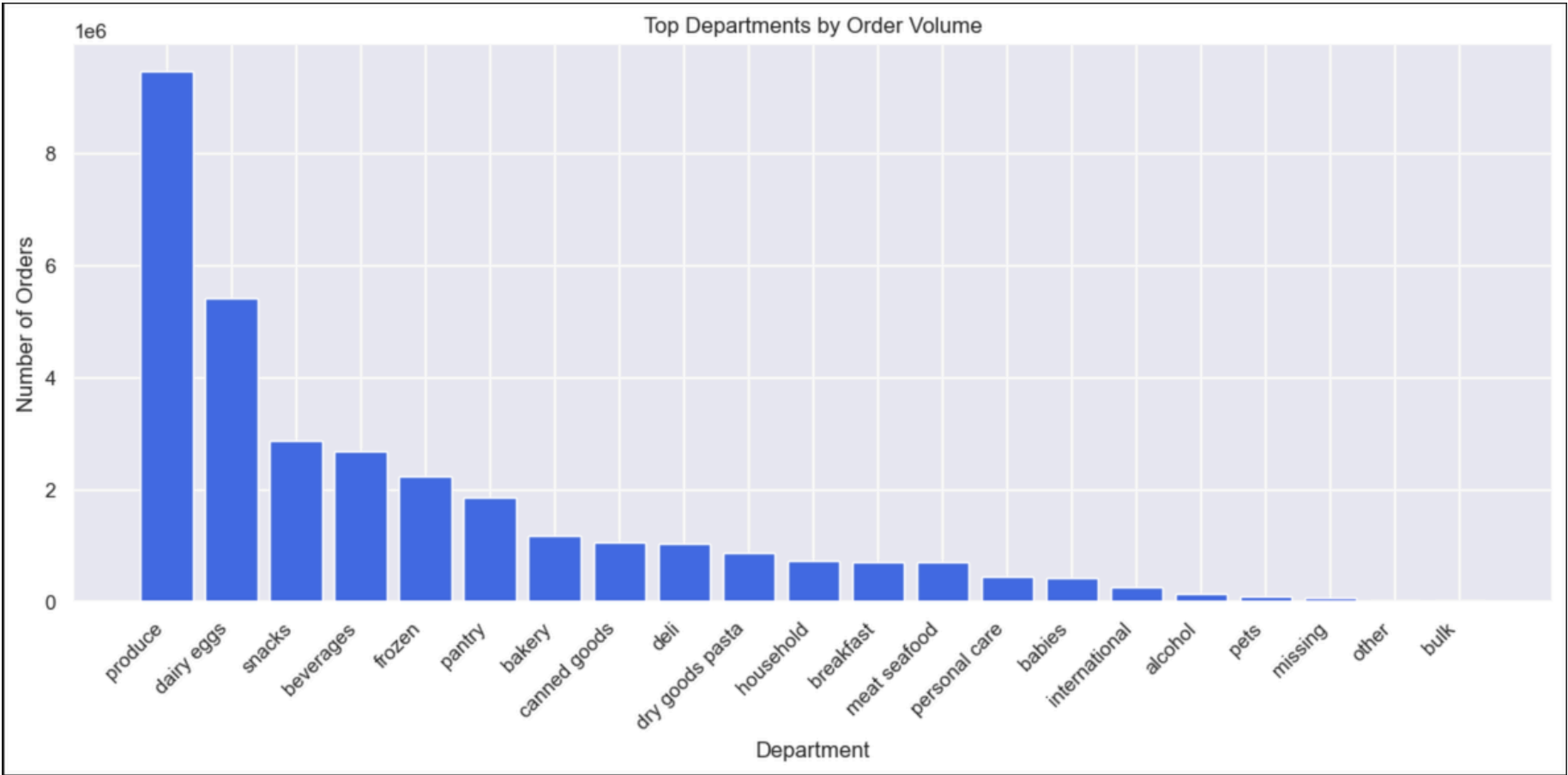


- The goal was to analyze over 3.4 million orders from 206,000 customers to uncover purchasing patterns and provide data-backed business recommendations.
- Upon analysis of the data, multiple insights were gained.
- Analysis of data involved using multiple libraries to either clean up the data or visualise data to make it understandable to interpret.
- A key data gathered was that the average size of the basket was found to be approximately 10 items.

Fresh & Organic Produce Dominate



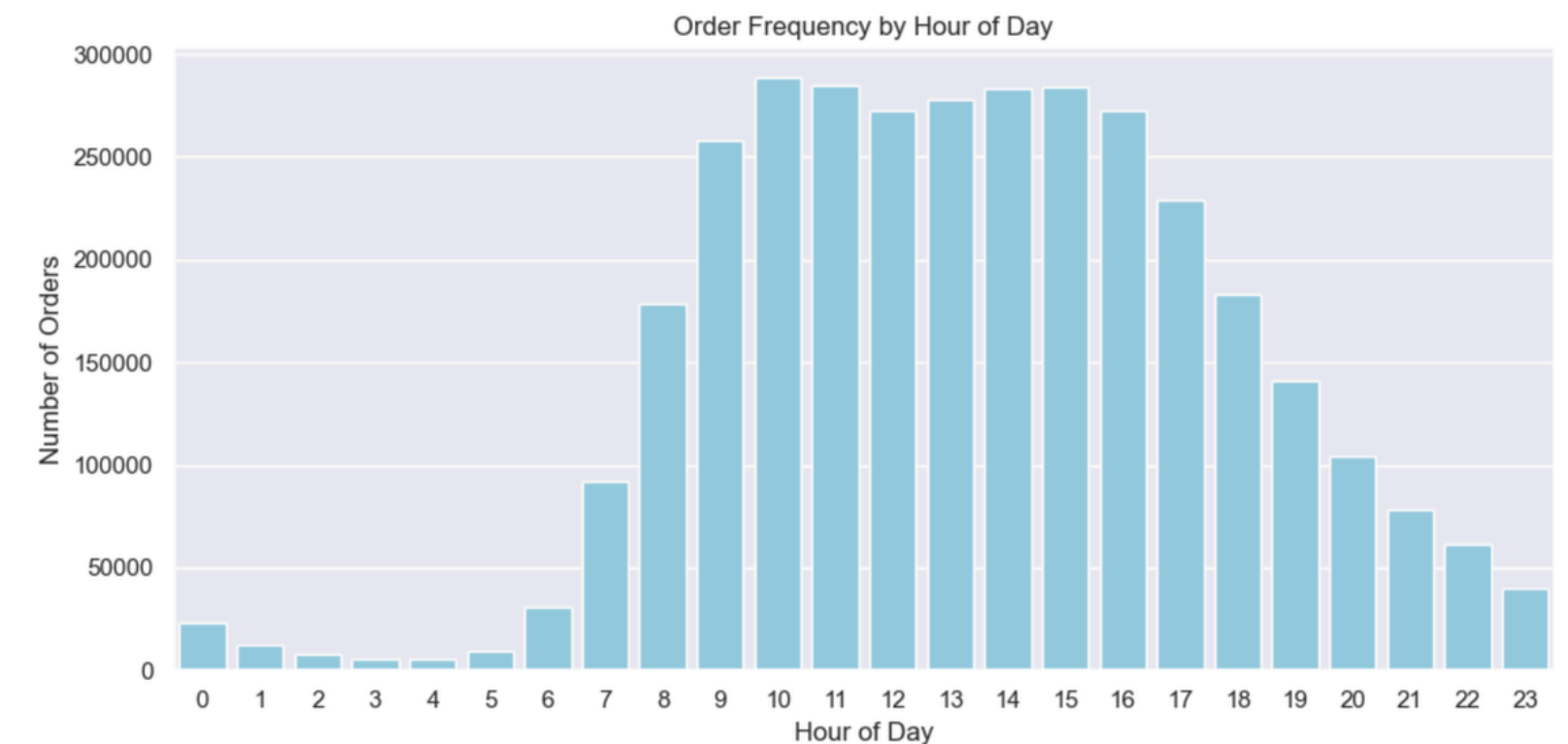
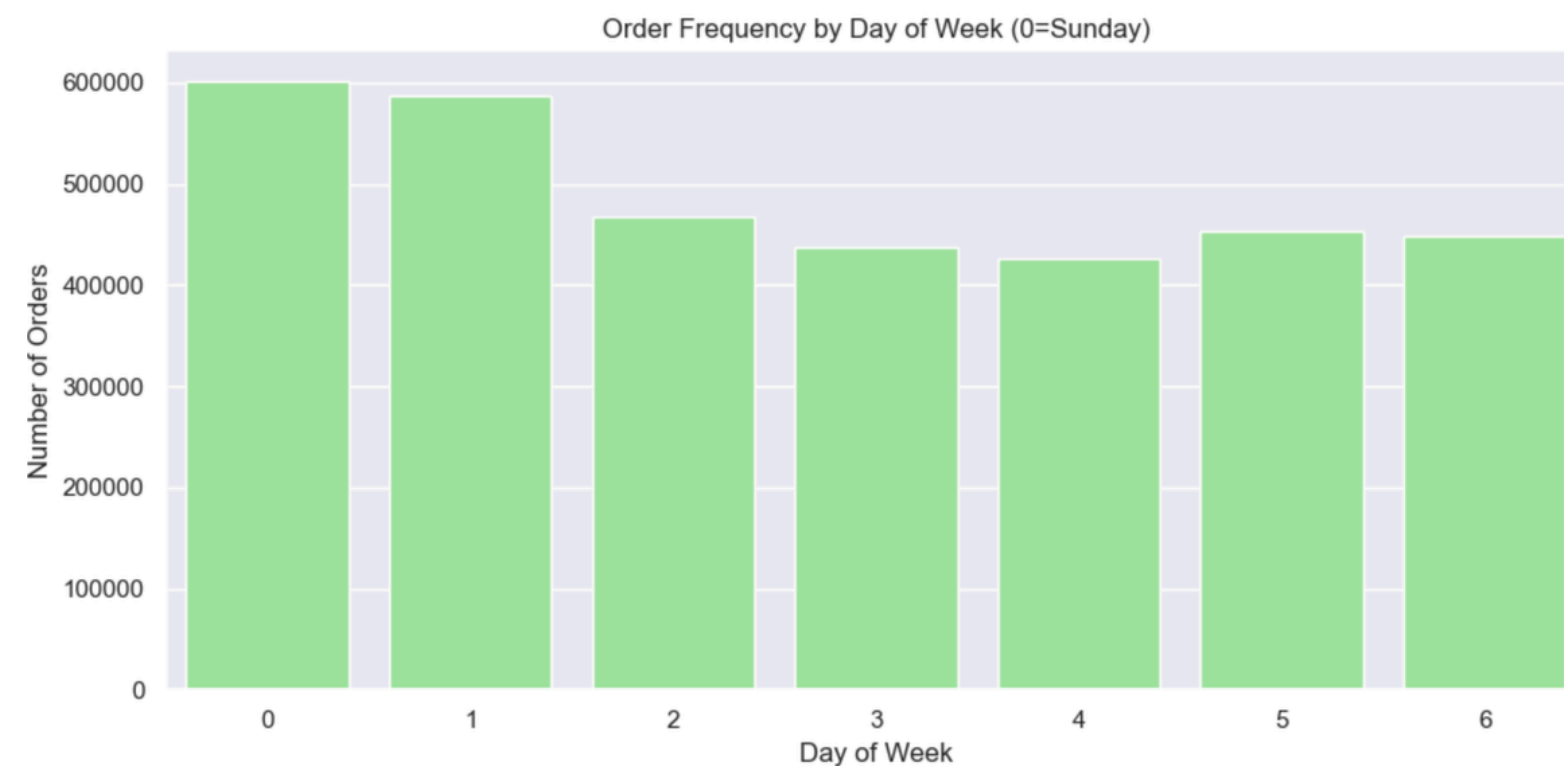
- The Produce and Dairy/Eggs departments drive the highest order volume.
- Few of the top selling items are : Bananas (both organic and regular), organic strawberries, and organic baby spinach are the top-selling items, highlighting a strong customer preference for fresh, organic options.



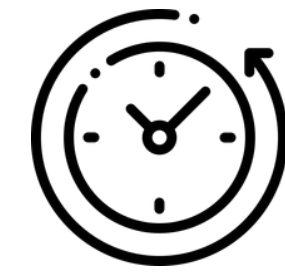
product_name	sales
Banana	472565
Bag of Organic Bananas	379450
Organic Strawberries	264683
Organic Baby Spinach	241921
Organic Hass Avocado	213584
Organic Avocado	176815
Large Lemon	152657
Strawberries	142951
Limes	140627
Organic Whole Milk	137905
Organic Raspberries	137057
Organic Yellow Onion	113426
Organic Garlic	109778
Organic Zucchini	104823
Organic Blueberries	100060
Cucumber Kirby	97315
Organic Fuji Apple	89632
Organic Lemon	87746
Apple Honeycrisp Organic	85020
Organic Grape Tomatoes	84255

Peak Ordering Times are Predictable

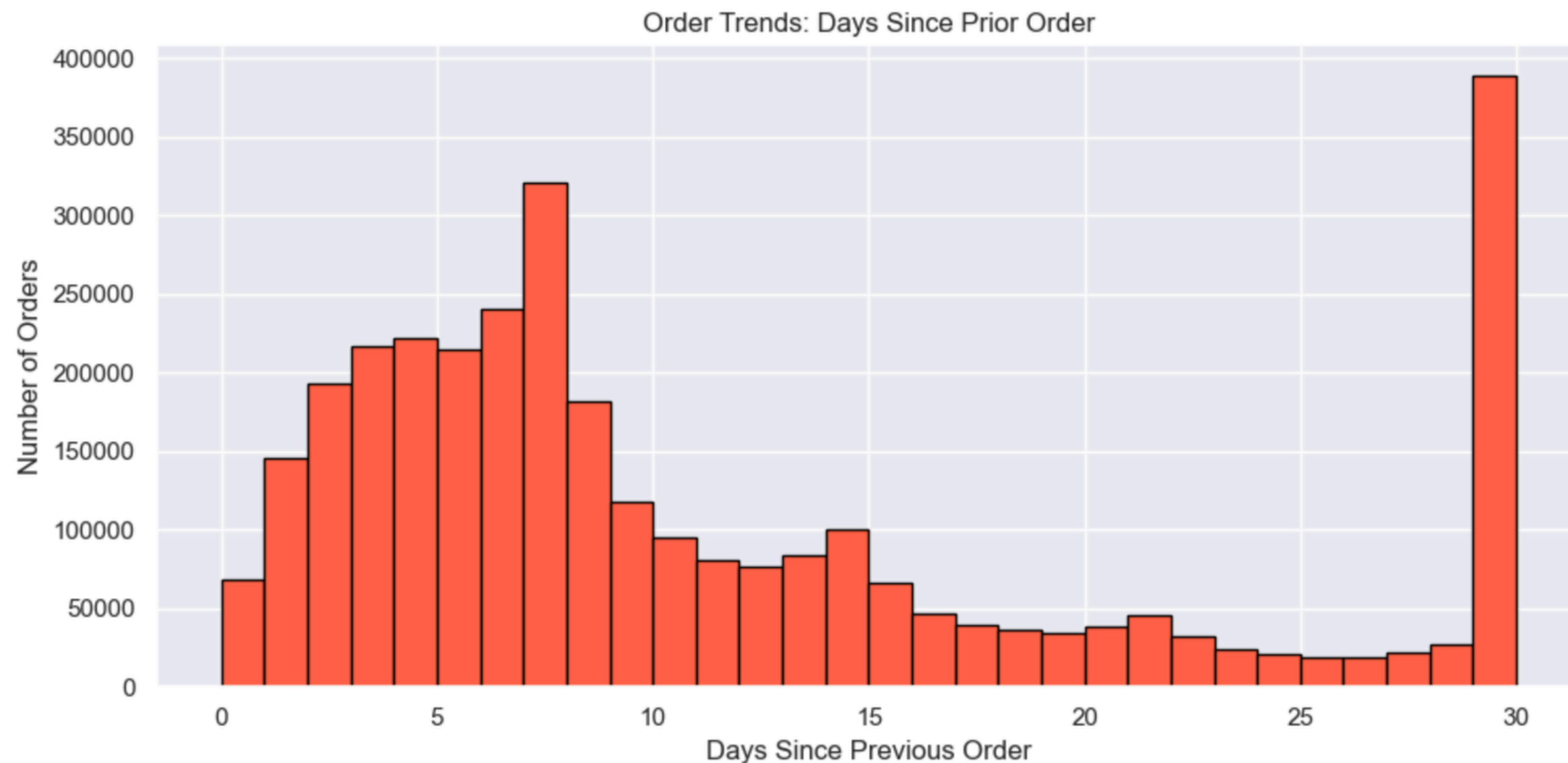
- The highest volume of orders occurs on Sundays and Mondays between 10AM & 3PM.
- We infer that customers tend to plan their weeks for grocery planning and shopping ahead during these windows.



Customers follow timely cycles



- The "Days Since Prior Order" data shows significant spikes at 7, 14, 21, and 30 days.
- This indicates strong weekly and monthly shopping habits, likely corresponding to weekly meal prep and monthly stock-ups.



Uncovering Product Pairings



- Market Basket Analysis reveals strong purchasing associations.
- The strongest pairings are found within the fresh produce category.

Support	How often the combo occurs
Confidence	Likelihood of consequent given antecedent
Lift	Degree of increase in buying above random

antecedents	consequents	support	confidence
(Organic Fuji Apple)	(Banana)	0.014436	0.378693
(Honeycrisp Apple)	(Banana)	0.012082	0.356128
(Cucumber Kirby)	(Banana)	0.013651	0.329826
(Organic Avocado)	(Banana)	0.022709	0.301982
(Seedless Red Grapes)	(Banana)	0.010460	0.297428
(Organic Raspberries)	(Bag of Organic Bananas)	0.017226	0.295519
(Organic Hass Avocado)	(Bag of Organic Bananas)	0.026514	0.291880
(Strawberries)	(Banana)	0.017536	0.288434
(Apple Honeycrisp Organic)	(Bag of Organic Bananas)	0.010078	0.278711
(Large Lemon)	(Banana)	0.017387	0.267790
(Organic Raspberries)	(Organic Strawberries)	0.014402	0.247072
(Organic Blueberries)	(Organic Strawberries)	0.010104	0.237418
(Organic Strawberries)	(Bag of Organic Bananas)	0.026211	0.232837
(Organic Yellow Onion)	(Bag of Organic Bananas)	0.010452	0.216652
(Organic Baby Spinach)	(Bag of Organic Bananas)	0.021424	0.208217
(Limes)	(Large Lemon)	0.011655	0.194863
(Organic Hass Avocado)	(Organic Strawberries)	0.017350	0.190997
(Organic Raspberries)	(Organic Hass Avocado)	0.010970	0.188192
(Large Lemon)	(Limes)	0.011655	0.179507
(Organic Avocado)	(Organic Baby Spinach)	0.013137	0.174697

=== Bundle Promotions (according to lift)===		
antecedents	consequents	lift
(Large Lemon)	(Limes)	3.001302
(Limes)	(Large Lemon)	3.001302
(Organic Strawberries)	(Organic Raspberries)	2.194801
(Organic Raspberries)	(Organic Strawberries)	2.194801
(Large Lemon)	(Organic Avocado)	2.126931
(Organic Avocado)	(Large Lemon)	2.126931
(Organic Blueberries)	(Organic Strawberries)	2.109035
(Organic Strawberries)	(Organic Blueberries)	2.109035
(Organic Hass Avocado)	(Organic Raspberries)	2.071710
(Organic Raspberries)	(Organic Hass Avocado)	2.071710
(Banana)	(Organic Fuji Apple)	1.884181
(Organic Fuji Apple)	(Banana)	1.884181
(Organic Raspberries)	(Bag of Organic Bananas)	1.831169
(Bag of Organic Bananas)	(Organic Raspberries)	1.831169
(Bag of Organic Bananas)	(Organic Hass Avocado)	1.808621
(Organic Hass Avocado)	(Bag of Organic Bananas)	1.808621
(Honeycrisp Apple)	(Banana)	1.771911
(Banana)	(Honeycrisp Apple)	1.771911
(Bag of Organic Bananas)	(Apple Honeycrisp Organic)	1.727016
(Apple Honeycrisp Organic)	(Bag of Organic Bananas)	1.727016

Inventory Suggestion : Stock the Winners



- We prioritize high demand products, the top 20 most bought products, which primarily include fresh and organic produce.

- Key Products include :

Banana	Organic Raspberries
Bag of Organic Bananas	Organic Yellow Onion
Organic Strawberries	Organic Garlic
Organic Baby Spinach	Organic Zucchini
Organic Hass Avocado	Organic Blueberries
Organic Avocado	Cucumber Kirby
Large Lemon	Organic Fuji Apple
Strawberries	Organic Lemon
Limes	Apple Honeycrisp Organic
Organic Whole Milk	Organic Grape Tomatoes

- We focus on top departments, i.e., the Produce and Dairy/Eggs departments are the most popular by a large margin. We ensure that these sections are well-stocked and visually appealing.

Smart Product Bundling: The "Perfect Pairings" Campaign

- Using the association rules, we are creating targeting bundles & “add on” suggestions. We are using the “lift” metric, as it shows the true influence one product's purchase has on another.
- Top 5 Bundles (Based on Highest Lift):
 - The Citrus Splash: Large Lemon + Limes (Lift: 3.00)
 - The Berry Boost: Organic Blueberries + Organic Strawberries (Lift: 2.19)
 - The Avocado & Lemon Duo: Organic Avocado + Large Lemon (Lift: 2.12)
 - The Organic Berry Mix: Organic Raspberries + Organic Hass Avocado (Lift: 2.07)
 - The Orchard Classic: Organic Fuji Apple + Banana (Lift: 1.88)
- Promotional Ideas:
 - In-Cart Pop-up: When a customer adds "Large Lemon" to their cart, prompt them:
“Complete the pair: Get limes for 10% off”
 - Homepage Banner: Feature "The Berry Boost" bundle with a catchy title.

Timed Promotional Notifications: Reach Customers at the Right Time



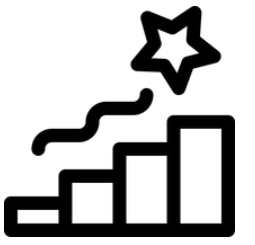
- We are trying to maximise engagement by sending promotions during peak shopping hours.
- We will schedule push notifications and promotional emails for Sundays and Mondays during 10 AM to 1 PM, being the busiest times on the app, to align with customers grocery planning and shopping preferences.

Hi Kshitij, got a few minutes to shop for yourself? Inbox x

FreshMart  <hello@freshmart.com> [Unsubscribe](#)
to me ▼

Sat, Aug 23, 12:49 PM

Featured Products & Homepage Strategy: Showcase What's Popular



- The homepage should be a direct reflection of what the customers love.
- Highlight Top Sellers: Create prominent banners for the absolute top-selling products like Bananas, Organic Strawberries, and Organic Avocados.
- Feature Popular Aisles: The Fresh Fruits and Fresh Vegetables aisles are the most visited. Create direct links or "Shop Now" buttons on the homepage that lead to these categories.
- Theme of the Week: Run a weekly theme based on the data. For example, a "Smoothie Essentials" week could feature bundles like avocados and spinach, or berries and bananas. Schedule push notifications and promotional emails for the campaign during the week.

Roadmap & Next Steps



Week 1:

- Update inventory levels for top products.
- Redesign and launch new homepage banners.
- Schedule the upcoming week's marketing notifications for the new peak times.

Weeks 2-3:

- Deploy the in-cart suggestion feature.
- Launch the "Perfect Pairings" pilot program with the top 3 bundles.

Week 4 & Beyond:

- Analyze the AOV and conversion data from the bundle pilot.
- If successful, expand the program to include the next 5-10 strongest product associations.
- Review notification engagement data and continue to refine send times.
- Refresh the analysis quarterly to identify new trends and update recommendations.

THANK YOU